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European Journal of Radiology

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About the journal

Aims and scope

European Journal of Radiology is an international journal which aims to communicate to its readers, state-of-the-art information on **imaging developments** in the form of high-quality original research articles and timely reviews on current developments in the field.

Its audience includes clinicians at all levels of training including **radiology** trainees, newly qualified **imaging** specialists and the experienced radiologist. Its aim is to inform efficient, appropriate and evidence-based imaging practice to the benefit of patients worldwide.

You are also welcome to submit to the *European Journal of Radiology*'s open access companion title *European Journal of Radiology Open* .

Article types

Non-commissioned articles include: • Research Papers (maximum 4,000 words including abstract, legends and up to 40 references) • Review articles (maximum 6,000 words including abstract and legends but excluding references and table content) • Short communications/Technical Notes (about 2,500 words including abstracts and legends but excluding references and table content) • Opinion Paper (about 3,000 words including abstracts and legends but excluding references and table content) • Letters to the Editor will be considered on controversial or technical issues • Please note that the *European Journal of Radiology* no longer publishes case reports.

Peer review

This journal follows a double anonymized review process. Your submission will initially be assessed by our editors. If your submission is deemed suitable, it will typically be sent to be reviewed by a minimum of two reviewers for an independent expert assessment of the scientific quality. Our editors decide whether to accept or reject your article.

Read more about peer review.

Our editors are not involved in making decisions about papers that:

- they have written themselves.
- have been written by family members or colleagues.

- relate to products or services in which they have an interest.

Any such submissions will be subject to the journal's usual procedures. Peer review will be handled independently of the editor involved and their research group. Read more about editor duties.

Authors may formally appeal the editorial decision, provided the appeal meets all the requirements and follows the procedure in Elsevier's Appeal Policy. Only one appeal per submission will be considered and the decision will be final.

Special issues and article collections

Special issues and article collections follow the same peer review process as regular submissions, except that a guest editor may send the submissions out to the reviewers and may recommend a decision to the journal editor. The journal editor oversees the peer review process of all special issues and article collections to ensure the journal's standards of publishing ethics and responsiveness, and is responsible for the final decision on acceptance or rejection of articles.

Open access

See our open access information page to learn about open access options for this journal.

Ethics and policies

Ethics in publishing

Authors must follow ethical guidelines stated in Elsevier's Publishing Ethics Policy.

Submission declaration

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- The article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.
- If accepted, the article will not be published elsewhere in the same form, in English or in any other language, including electronically, without the written consent of the copyright holder.

To verify compliance with our journal publishing policies, we may check your manuscript with our screening tools.

Authorship

All authors should have made substantial contributions to all of the following:

1. The conception and design of the study, or acquisition of data, or analysis and interpretation of data.
2. Drafting the article or revising it critically for important intellectual content.
3. Final approval of the version to be submitted.

All authors are accountable for all aspects of the work, to ensure that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Corresponding author

Authors must appoint a single corresponding author to communicate with the journal during the editorial process. This individual submits the manuscript and all required documentation to the journal, and serves as the primary contact between the journal and the co-authors. It is important that all contact details of the corresponding author are kept up to date throughout the submission and publication process.

Only the corresponding author's affiliation will be used to determine eligibility for a publishing agreement and possible discounts related to it. Affiliations of other co-authors are not relevant for eligibility.

Group authorship

Group authorship refers to the attribution of an article to a defined and named group of authors whose members have contributed to the work. Examples include research consortia, clinical trial groups, collaborative networks, task forces or working groups, expert panels or consensus groups, and participatory research groups.

- Group authorship must be declared on the title page *at submission*. When available, the submitting author should include the group under "Collaborative Author Groups" in the submission system.
- A corresponding author who agrees to coordinate all communications on behalf of the group must be designated.
- The group name will be included in the author list in the published article (e.g., "on behalf of the [Consortium Name]").
- All group members associated with the work should be listed in a dedicated section, preferably an appendix.
- All listed authors must follow the journal's authorship and authorship change policies.
- The group must maintain a transparent internal process for determining authorship and resolving disputes. All authorship disputes must be resolved within the group.

Changes to authorship

The editors of this journal generally will not consider changes to authorship once a manuscript has been submitted. It is important that authors carefully consider the authorship list and order of authors and provide a definitive author list at submission.

This journal's policy on authorship changes:

- All authors must be listed in the manuscript and their details entered into the submission system. Changes can only be made prior to acceptance, and only if approved by the journal editor. This includes addition, deletion, or rearrangement of author names.
- Requests to change authorship must be made by the corresponding author using the Authorship Change Request form. The corresponding author must provide the reason for the request to the journal editor with written confirmation from all authors, including any authors being added or removed, that they agree with the changes. Requests that do not comply with the instructions outlined in the form will not be considered.
- This journal does not allow authorship changes after acceptance. This includes addition, deletion, or rearrangement of author names, and changes to the corresponding author.
- The review process may be paused while an authorship change request is being considered.
- Change requests made by the authors without the Authorship Change Request form or editor approval may result in the rejection of the manuscript, or retraction if the article has already been published.

Declaration of competing interests

All authors must disclose any financial and personal relationships with other people or organizations that could inappropriately influence or bias their work. Examples of potential competing interests include:

- Employment
- Consultancies
- Stock ownership
- Honoraria
- Paid expert testimony
- Patent applications or registrations
- Grants or any other funding

- Affiliation with the journal as an editor or advisory board member

The declarations tool should always be completed.

Authors with a journal affiliation to declare should enter the following text under “Other Activities” within the declarations tool and should inform the journal and publisher prior to completing the submission process:

Given their role as [AUTHOR JOURNAL ROLE TITLE], [AUTHOR NAME] had no involvement in the peer review of this article and has no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor.

Editorial disclosure statements will be included as a footnote and/or in the declaration of competing interests section of the article.

Authors with no competing interests to declare should select the option "I have nothing to declare."

The resulting Word document containing your declaration should be uploaded at the "attach/upload files" step in the submission process. Save the document in .doc/.docx format. Author signatures are not required.

Funding sources

Authors must disclose any funding sources that provided financial support for the conduct of the research and/or preparation of the article. The role of sponsors, if any, should be declared in relation to the study design, collection, analysis, and interpretation of data, writing of the report, and decision to submit the article for publication. If funding sources had no such involvement, this should be stated in your submission.

List funding sources in this standard way to facilitate compliance with funder's requirements:

Funding: This work was supported by the National Institutes of Health [grant numbers xxxx, yyyy]; the Bill & Melinda Gates Foundation, Seattle, WA [grant number zzzz]; and the United States Institutes of Peace [grant number aaaa].

It is not necessary to include detailed descriptions of the program or type of grants, scholarships and awards. When funding is from a block grant or other resources available to a university, college, or other research institution, submit the name of the institute or organization that provided the funding.

If no funding has been provided for the research, please include the following sentence:

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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Authors must declare the use of generative AI tools in the manuscript preparation process upon submission of the paper.

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- Ensuring accuracy and originality of any AI generated images and providing appropriate attribution where relevant.
- Ensuring the use of any tools or sources, AI based or otherwise, is made clear and transparent to readers. If AI tools have been used a disclosure statement is required upon submission, example below.

- Ensuring the manuscript is developed in a way that safeguards data privacy, intellectual property, and other rights, by checking the terms and conditions of any AI tool that is used.

Finally, authors must not list or cite AI tools as an author or co-author on the manuscript since authorship implies responsibilities and tasks that can only be attributed to, and performed by, humans.

The use of AI tools in the manuscript preparation process must be declared by adding a statement at the end of the manuscript when the paper is first submitted. The statement will appear in the published work and should be placed in a new section before the reference list. An example:

Title of new section: *Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.*
Statement: *During the preparation of this work the author(s) used [NAME OF TOOL/SERVICE] to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.*

The declaration does not apply to the use of basic tools, such as tools used to check grammar, spelling, and references. In addition, authors are not required to disclose the use of AI tools or features within specialist disability-related assistive technology, provided these are used solely for accessibility purposes.

If you have nothing to disclose, you do not need to add a statement.

Please note: to protect authors' rights and the confidentiality of their research, this journal does not currently allow academic editors and reviewers to upload an unpublished manuscript into a generative AI tool, as stated in Elsevier's GenAI policies for journals. However, AI tools may be used in a supportive capacity by editors and reviewers to improve structure and language of the review report or editorial decision letter or to conduct a background literature search. Elsevier is actively evaluating compliant AI tools and may revise this policy in the future.

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Preprint sharing

Authors may share preprints in line with Elsevier's article sharing policy. Sharing preprints, such as on a preprint server, will not count as prior publication.

We advise you to read our policy on multiple, redundant, or concurrent publication.

Use of inclusive language

Inclusive language acknowledges diversity, conveys respect to all people, is sensitive to differences, and promotes equal opportunities. Authors should ensure their work uses inclusive language throughout and contains nothing that might imply one individual is superior to another on the grounds of:

- age
- gender
- race
- ethnicity
- culture
- sexual orientation
- disability or health condition

Avoid the use of descriptors about personal attributes unless they are relevant and valid. Write for gender neutrality with the use of plural nouns ("clinicians, patients/clients") as default. Wherever possible, avoid using "he, she," or "he/she."

Don't make assumptions about the beliefs of readers and keep writing free from bias, stereotypes, slang, reference to dominant culture, and cultural assumptions.

These guidelines are meant as a point of reference to help you identify appropriate language but are by no means exhaustive or definitive.

Reporting sex- and gender-based analyses

There is no single, universally agreed-upon set of guidelines for defining sex and gender. We offer the following guidance:

- Sex- and gender-based analyses (SGBA) should be integrated into research design when research involves or pertains to humans, animals, or eukaryotic cells. This should be done in accordance with any requirements set by funders or sponsors and best practices within a field.
- Sex and/or gender dimensions of the research should be addressed within the article or declared as a limitation to the generalizability of the research.
- Explicitly state applied definitions of sex and/or gender. This enhances precision, rigor, and reproducibility of the research, and avoids confusing terms and the constructs they refer to.

We advise you to read the Sex and Gender Equity in Research (SAGER) guidelines and the SAGER checklist (PDF) on the EASE website, which offer systematic approaches to the use of sex and gender information in study design, data analysis, outcome reporting, and research interpretation.

For further information we suggest reading the rationale behind and recommended use of the SAGER guidelines.

Definitions of sex and/or gender

We ask authors to define how sex and gender have been used in their research and publication. Some guidance:

- Sex generally refers to a set of biological attributes that are associated with physical and physiological features such as chromosomal genotype, hormonal levels, and internal and external anatomy. A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth") and is in most cases based solely on the visible external anatomy of a newborn. In reality, sex categorizations include people who are intersex/have differences of sex development (DSD).
- Gender generally refers to socially constructed roles, behaviors, and identities of women, men, and gender-diverse people that occur in a historical and cultural context and may vary across societies and over time. Gender influences how people view themselves and each other, how they behave and interact, and how power is distributed in society.

Depending on the focus of a paper, sex and/or gender may or may not be relevant to the content of the paper. We recognize that beliefs, attitudes, and laws relating to sex and gender may vary. These articles do not attempt to dictate author beliefs but rather require that, where relevant to an author's research or paper, the author must provide clear explanations of how the paper and research define and use sex and gender.

Image manipulation

We accept that authors sometimes need to adjust images for clarity, but any manipulation of images for the purpose of deception or fraud will be seen as scientific ethical abuse and will be dealt with accordingly.

Authors must adhere to this journal's policy for graphical images:

- No specific feature within an image may be enhanced, obscured, moved, removed, or introduced.
- Adjustments of brightness, contrast, or color balance are acceptable as long as they do not obscure or eliminate any information present in the original image.
- Nonlinear adjustments such as changes to gamma settings must be disclosed in the figure legend.
- Use of AI tools to create or manipulate primary research images is not permitted. This includes images that represent experimental or observational data, such as western blots, microscopy images, histology images, or patient images. Please read our policy on the use of AI tools in figures, images, and artwork, which can be found in Elsevier's GenAI Policies for Journals.

To verify compliance with the above, this journal may send your images to a secure third-party tool to enable screening for image irregularities. Our editors make the final decision on whether images are acceptable, and they may ask you to provide original data or images if any questions arise as a result of the screening.

Authors are encouraged to carefully check all images before submission and to connect all the data in any figures to the original, unprocessed data.

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- Maps: Readers should be able to locate any study areas shown within maps using common mapping platforms. Maps should only show the area actually studied, and authors should not include a location map which displays a larger area than the bounding box of the study area. Authors should add a note clearly stating that "*map lines delineate study areas and do not necessarily depict accepted national boundaries.*" Editors may request authors to change maps during the review process if these guidelines are not followed.

- Institutional affiliations: Authors should use either the full, standard title of their institution or the standard abbreviation of the institutional name so that the institutional name can be independently verified for research integrity purposes.

Studies in humans

Authors must follow ethical guidelines for research involving humans, their data, and/or their biological materials (e.g., organs, tissues). Authors should ensure that they adhere to the following:

- All studies must be performed in compliance with relevant laws, regulations, and must be approved by the appropriate institutional committee(s). For instance, medical research involving humans should be carried out in accordance with the World Medical Association Declaration of Helsinki: Ethical principles for medical research involving human participants and should follow the International Committee of Medical Journal Editors (ICMJE) recommendations for the conduct, reporting, editing and publication of scholarly work in medical journals and aim to be representative of human populations in terms of sex, age, and ethnicity. Sex and gender terms should be used correctly, as outlined in the "Reporting sex- and gender-based analyses" section of this guide for authors.

- Informed consent must be obtained for research with human participants, including their data or biological materials. Consent must be given without coercion, and the privacy rights of human participants must always be observed. If your study contains human participants, please read Elsevier's full human research policy for further information.

    - Authors should be aware that written (or electronic) consent is required for certain studies under national laws. In studies where only verbal informed consent was obtained, authors should include justification for the absence of written consent, the name of the approving ethics committee, and a statement on how verbal consent was recorded.

    - Consent for publication should also be obtained.

- Authors must obtain and retain explicit written consent for any identifiable participant materials (e.g., case details, videos, recordings, images, photographs, and illustrations). All participant information must be de-identified: do not include names or personal identifiers (e.g., initials, ID numbers, DOB)

- Images should be used only when scientifically necessary and with explicit consent, and nonessential identifying details must be removed. If identifiers are altered to ensure anonymity, authors must confirm that scientific meaning is not affected. Fully anonymized images (e.g., x-rays, scans) that contain no identifying marks and are not accompanied by identifying text do not require formal consent, though informing the patient is good practice. Case reports always require consent. Simply obscuring identity (e.g., eye bars or blurring) is not sufficient without consent.

- Written consents must be retained by the authors. They should not be provided to this journal unless this is specifically requested in exceptional circumstances, for example, when a legal issue arises. Only then should you provide copies of the consents, or evidence that all relevant consents were obtained.

- Appropriate consents and safeguards should be in place for research with vulnerable populations, as described in Elsevier's human research policy. In many countries, additional ethical approval may be needed for research with those who lack the capacity to consent (e.g., children or those with mental health or neurological conditions including dementia).

- For the full policy, see "Consent requirements for case reports, personal information, and images."

All studies involving human participants require a statement that details any ethical approval(s), guidelines followed, confirmation of informed consent, and any other necessary information in line with Elsevier's human research policy. The statement should contain the date and reference number of any ethical approval(s) obtained.

If ethical approval is not required by national laws or where a study has been granted an exemption from an ethics committee, the statement must explain why the study is exempt and confirm that the appropriate protocols for protecting the rights and privacy of all participants were utilized during the execution of the research (e.g., no coercion to participate, full disclosure of study requirements and risks, consent of participants (or their legal representatives), no release of participant data without their knowledge, etc.).

Human biological material

For all studies that use human organs or tissues, sufficient evidence must be provided that these were procured in line with WHO Guiding Principles on Human Cell, Tissue and Organ Transplantation. The source of the organs or tissues used in clinical research must be transparent and traceable and must not be sourced from executed prisoners or prisoners of conscience, consistent with recommendations by Global Rights Compliance on Mitigating Human Rights Risks in Transplantation Medicine.

For clinical studies, a statement that informed consent was obtained from the patient or their legal representative, together with ethical approval from a suitable institution, may be sufficient evidence. The journal has the right to request additional evidence in cases where it feels this is needed.

If the manuscript describes organ transplantation, authors must additionally declare within the manuscript that:

- Autonomous consent, free from coercion, was obtained from the donor(s) or their next of kin.
- Organs and/or tissues were not sourced from executed prisoners or prisoners of conscience.

Studies in animals

Authors must follow ethical guidelines for studies carried out in animals.

All animal experiments should comply with the ARRIVE (Animal Research: Reporting of In Vivo Experiments) guidelines (or national equivalent) and should be carried out in accordance with relevant laws and associated guidelines for animal welfare. This may include the U.K. Animals (Scientific Procedures) Act 1986; the EU Directive 2010/63/EU on the protection of animals used for scientific purposes; the National Research Council Guide for the Care and Use of Laboratory Animals; or those of an equivalent internationally recognized body.

Authors must ensure that the manuscript contains a statement that confirms which guidelines have been followed. This statement should include the ethics committee approval number, when available. If a study was granted exemption or did not require ethics approval, a statement detailing this should be included in the manuscript.

The sex of animals and, where appropriate, the influence (or association) of sex on the results of the study should be indicated.

Where applicable, researchers are encouraged to document informed consent from the client or owner in studies involving client-owned animals and adhere to best practices of veterinary care.

For research on threatened and/or endangered species, authors are advised to consult international best practice guidelines such as the International Union for Conservation of Nature (IUCN) Policy Statement on Research Involving Species at Risk of Extinction and IUCN Red List guidelines.

Clinical Trials

This journal follows the International Committee of Medical Journal Editors (ICMJE) clinical trials registration guidelines.

Registration

Clinical trials must be registered in a public trials registry. Purely observational studies, in which the assignment of the medical intervention is not at the discretion of the investigator, do not require registration.

Some important points:

- Trials must be registered at or before the onset of patient enrollment.
- The clinical trial registration number must be included in the manuscript, preferably at the end of the article abstract.
- A clinical trial is defined as any research study that prospectively assigns human participants, or groups of humans, to one or more health-related interventions to evaluate the effects on health outcomes.

- Health-related interventions include any intervention used to modify a biomedical or health-related outcome such as drugs, surgical procedures, devices, behavioral treatments, dietary interventions, and process-of-care changes.

- Health outcomes include any biomedical or health-related measures obtained in patients or participants, including pharmacokinetic measures and adverse events.

Results

Authors must disclose all posting in registries of results of the same or closely related work. Editors will not consider results to be a prior publication if they have already been posted in the same clinical trials registry in which primary registration resides, as long as the results are presented in the form of a brief structured abstract (fewer than 500 words) or table.

Disclosing results in other circumstances, for example at an investors' meeting, is discouraged and may jeopardize consideration of your manuscript by this journal.

Reporting

Authors are encouraged to follow CONSORT guidelines when presenting randomized controlled trials. Provide the CONSORT checklist at manuscript submission, with an accompanying flow diagram illustrating the progress of patients through the trial, including recruitment, enrollment, randomization, withdrawal, completion, and a description of the randomization procedure.

- Read the CONSORT guidelines

- Follow the CONSORT checklist

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If you need to delay the publication of your article for any reason, please contact the editorial office of the journal or our Journal Article Publishing Support Center at the earliest possible opportunity. While we will try to ensure that your article is not published before a certain date if you make your request on a timely basis, we cannot guarantee this. Please also note that once the agreed period of delay has expired the article may be published at any time thereafter.

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For questions about article withdrawal, correction, or retraction, please consult Elsevier's Article Correction, Retraction and Removal Policy.

Writing and formatting

File format

Submit your manuscript and additional text-based files in an editable format:

- Submit files with .doc/.docx extensions for Word and .tex for LaTeX files. A PDF is not an acceptable source file.

- Format Word files in a single-column layout. Double-column formatting is only permitted for LaTeX submissions.

- Remove all strikethrough and underline formatting, unless it is scientifically relevant.

Title page

You are required to include the following details on the title page:

- Article title. Article titles should be concise and informative. Please avoid abbreviations and formulae, where possible, unless they are established and widely understood, e.g., DNA.

- Author names. Provide the given name(s) and family name(s) of each author. The order of authors should match the order in the submission system. Carefully check that all names are accurately spelled. You may add your name in parentheses in your own script after the English transliteration.

- Affiliations, referring to where the work was carried out, below the author names. Indicate affiliations using a lowercase superscript letter immediately after the author's name and in front of the corresponding address. Ensure that you provide the full name and postal address of each affiliation, including the country name.

- Corresponding author. Clearly indicate who will handle correspondence for your article: during the peer review and publication process, and also after publication. This responsibility includes answering any future queries about your results, data, or methodology. The email address of the corresponding author must be included on the title page and will be published in the article.

- Present/permanent address. If an author has changed affiliation since the work described in your article was carried out, or was visiting during that time, a "present address" (or "permanent address") can be indicated by a footnote to the author's name. The address where the author carried out the work must be retained as their main affiliation address. Use superscript Arabic numerals for such footnotes.

Double anonymized peer review

This journal follows a double anonymized review process, meaning author identities are concealed from reviewers and vice versa.

The title page (including author details) and anonymized manuscript (excluding author details) need to be submitted as separate files. In addition to the items mentioned above, the title page should also include:

- Acknowledgments. Include any individuals who provided you with help during your research, such as help with language, writing, or proofreading in the acknowledgments section. Include acknowledgments only in the title page, do not add them as a footnote to your title, or anywhere else in your article.

- Declaration of competing interests (when a separate declaration of interest file is not submitted).

- Corresponding author address: full address is required along with the email address.

The anonymized manuscript should contain the main body of your paper, including references and tables. It is important that your manuscript and any supplementary materials do not contain any identifying information such as author names or affiliations, or acknowledgments.

Read more about peer review.

Abstract

You are required to provide a concise and factual abstract that does not exceed 250 words. The abstract should:

- State the purpose of your paper.

- Outline basic procedures followed.

- Include your main findings, their statistical significance, where applicable, and your principal conclusions.

- Emphasize new and important aspects of your study or observations.

Some guidelines:

- Abstracts must be able to stand alone, as abstracts are often presented separately from the article.

- Avoid references. If any are essential to include, ensure that you cite the author(s) and year(s).

- Avoid non-standard or uncommon abbreviations. If any are essential to include, ensure they are defined within your abstract at first mention.

Keywords

You are required to provide 1 to 7 keywords for indexing purposes. Keywords should be written in English. Try to avoid keywords consisting of multiple words (using "and" or "of").

Only use abbreviations in keywords if they are firmly established in the field.

Highlights

You are required to provide article highlights at submission.

Highlights are a short collection of bullet points that should capture the novel results of your research as well as any new methods used during your study. Highlights will help increase the discoverability of your article via search engines. Some guidelines:

- Submit highlights as a separate editable file in the online submission system with the word "highlights" included in the file name.
- Highlights should consist of 3 to 5 bullet points, each a maximum of 85 characters, including spaces.

We encourage you to view example article highlights and read about the benefits of their inclusion.

Graphical abstract

You are encouraged to provide a graphical abstract at submission.

The graphical abstract summarizes the contents of your article in a concise, pictorial, professional form, highlighting important aspects of the research. It is designed to appeal to an interdisciplinary audience and draw more attention to your online article. Some guidelines:

- Ensure you have obtained the necessary permission to include any third-party-owned material in a graphical abstract.
- Use of generative AI or AI-assisted tools in the production of graphical abstracts must align with Elsevier's GenAI Policies for Journals.
- Submit your graphical abstract as a separate file in the online submission system.
- Ensure the image is at least 1328 pixels wide, with the longest side horizontal, and is readable at a width of 13 cm.
- Preferred file types for graphical abstracts are TIFF, EPS, PDF, or MS Office files.

More information, examples, and a template can be found in the graphical abstract guidelines.

Units, classifications codes and nomenclature

Units

This journal requires you to use the International System of Units (SI) which follows internationally accepted rules and conventions. If other units are mentioned within your article, you should provide the equivalent unit in SI.

Tables

Tables must be submitted as editable text, not as images. Some guidelines:

- Place tables next to the relevant text or on separate page(s) at the end of your article.
- Cite all tables in the manuscript text.
- Number tables consecutively according to their appearance in the text.
- Please provide captions along with the tables.
- Place any table notes below the table body.
- Avoid vertical rules and shading within table cells.

Use tables sparingly, ensuring that any data presented in tables is not duplicating results described elsewhere in the article.

Figures, images, and other artwork

Figures, images, diagrams, and other graphical media (artwork) must be supplied as separate files along with the manuscript, in line with these artwork and media instructions. Some excerpts:

When submitting artwork:

- Cite all images in the manuscript text.
- Number images according to the order they appear within your article.

- Submit each image as a separate file using a logical naming convention for your files (for example, Figure_1, Figure_2 etc.).
- Text graphics may be embedded in the text at the appropriate position. If you are working with LaTeX, text graphics may also be embedded in the file.

Captions

All artwork must have a caption. A caption should consist of a brief title (not displayed on the figure itself) and a description of the image. Keep the amount of text to a minimum, though any symbol or abbreviation should be explained.

More information about adding captions to your submission..

Formats

Submit your artwork in the formats listed below, following the given requirements:

Image type	Recommended file type	Requirements
Vector drawings	EPS, PDF	Embed the font or save text as graphics
Photographs (color or grayscale)	TIFF, JPG, PNG	Minimum resolution 300 dpi (min. width: single column: 1063 pixels; full page: 2244 pixels)
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Research data

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[1] J. van der Geer, T. Handgraaf, R.A. Lupton, The art of writing a scientific article, J. Sci. Commun. 163 (2020) 51 – 59. https://doi.org/10.1016/j.sc.2020.00372.

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[3] W. Strunk Jr., E.B. White, The Elements of Style, fourth ed., Longman, New York, 2000.

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[4] G.R. Mettam, L.B. Adams, How to prepare an electronic version of your article, in: B.S. Jones, R.Z. Smith (Eds.), Introduction to the Electronic Age, E-Publishing Inc., New York, 2020, pp. 281 - 304.

Reference to a website:
[5] Cancer Research UK, Cancer statistics reports for the UK. http://www.cancerresearchuk.org/aboutcancer/statistics/cancerstatsreport/, 2023 (accessed 13 March 2023).

Reference to a dataset:
[6] M. Oguro, S. Imahiro, S. Saito, T. Nakashizuka, Mortality data for Japanese oak wilt disease and surrounding forest compositions [dataset], Mendeley Data, v1, 2015. https://doi.org/10.1234/abc12nb39r.1.

Reference to software:
[7] E. Coon, M. Berndt, A. Jan, D. Svyatsky, A. Atchley, E. Kikinzon, D. Harp, G. Manzini, E. Shelef, K. Lipnikov, R. Garimella, C. Xu, D. Moulton, S. Karra, S. Painter, E. Jafarov, S. Molins, Advanced Terrestrial Simulator (ATS) v0.88 [software], Zenodo, March 25, 2020. https://doi.org/10.1234/zenodo.3727209.

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